# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 3 – Comparative Analysis

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|---|---|---|---|
| Course Code: | ITA05 | Course Title: | Computer Vision |
| CO Assessed: | CO4 – Articulation (combining methods for evaluation) P4 | Assessment: | Assessment Tool 3 – Comparative Analysis |
| Weightage: | 10% of CIA | Total Marks: | 100 |
| CO4 Statement: | *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)* | | |
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| Section / Batch: | Section A | Date: | 30 August 2026 |

**Code of Conduct**

*I ______________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the Student: ____________________________*

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning | Marks |
|---|---|---|---|---|---|---|
| Scope of Work | 20% | Covers topic in detail with wide scope and justification | Covers appropriate scope with minor gaps | Limited to basic scope; lacks depth | Scope too narrow or not relevant | |
| Comparison Depth | 25% | Provides detailed and comprehensive comparison across multiple aspects | Provides adequate comparison with most aspects | Limited comparison with few aspects | Very minimal or no comparison | |
| Use of Evidence | 20% | Supports comparison with relevant data, examples, or references | Uses some appropriate evidence with minor gaps | Limited or weak evidence support | No supporting evidence provided | |
| Critical Thinking | 20% | Demonstrates strong critical thinking with clear interpretation and reasoning | Shows reasonable interpretation with minor gaps | Basic interpretation lacks depth | No meaningful interpretation | |
| Clarity of Presentation | 10% | Well-structured, clear, and logically organized presentation | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand | |
| Professionalism | 5% | Highly professional, well-documented, and clearly presented work | Good documentation and presentation | Basic documentation with some gaps | Poor documentation and presentation | |

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## 7. F1-Score Comparison

**Given:**

| **Model** | **Precision** | **Recall** |
|---|---|---|
| A | 0.80 | 0.60 |
| B | 0.75 | 0.75 |

**(a) Calculate F1-score**

The F1-score is the harmonic mean of precision and recall. It is useful when we want to evaluate both the correctness of detections and the ability to find relevant objects.

$F1{=}Precision{+}Recall2{\times}Precision{\times}Recall$

Model A

$F1A{=}0.80{+}0.602(0.80)(0.60)$

$=1.400.96$

$=0.686$

Therefore,

F1-score of Model A = 0.686 ≈ 68.6%

Model B

$F1B{=}0.75{+}0.752(0.75)(0.75)$

$=1.501.125$

$=0.75$

Therefore,

F1-score of Model B = 0.75 = 75%

**(b) Comparison and Evaluation**

| Parameter | Model A | Model B |
|---|---|---|
| Precision | 0.80 | 0.75 |
| Recall | 0.60 | 0.75 |
| F1-score | 0.686 | 0.750 |

Model A has a higher precision of 0.80, which means that when it predicts an object, its predictions are relatively accurate. However, its recall is only 0.60, meaning that it misses a greater proportion of relevant objects.

Model B has a slightly lower precision of 0.75, but its recall is also 0.75. This indicates a more balanced performance between finding relevant objects and avoiding incorrect detections.

The F1-score confirms this balance. Model B achieves 0.75, compared with 0.686 for Model A. The difference is:

0.75−0.686=0.064

Thus, Model B improves the F1-score by approximately 6.4 percentage points.

**Conclusion**

Model B is the better overall model because it achieves the higher F1-score of 0.75. Although Model A has better precision, Model B provides a better balance between precision and recall. Therefore, for a computer vision detection system where both missed detections and incorrect detections are important, Model B is preferable.

## 8. Optical Flow Consistency

Given motion values:

| Frame | Method A | Method B |
|---|---|---|
| 1 | 4 | 6 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

Optical flow is used in computer vision to estimate the motion of objects or pixels between consecutive frames. A stable method should produce motion estimates that do not fluctuate excessively from one frame to another.

**(a) Compute Average Motion**

The average motion is calculated as:

*Average*=Number of framesSum of motion values

Method A

*AverageA*=34+5+6

=315=5

Therefore,

Average motion of Method A = 5

Method B

*AverageB*=36+8+9=323

=7.67

Therefore,

Average motion of Method B = 7.67

**(b) Compare Consistency and Stability**

To determine which method is more stable, we examine the variation of motion values across the frames.

Range

The range is:

$Range=Maximum-Minimum$

For Method A:

$RangeA=6-4=2$

For Method B:

$RangeB=9-6=3$

Method A has a smaller range, indicating that its motion estimates vary less.

Variance

Using population variance:

$Variance=n\sum(x-\bar{x})2$

For Method A, the mean is 5:

$VarianceA=3(4-5)2+(5-5)2+(6-5)2$

$=31+0+1$

$=0.67$

For Method B, the mean is 7.67:

$VarianceB\approx3(6-7.67)2+(8-7.67)2+(9-7.67)2\approx1.56$

**Comparison**

| Measure | Method A | Method B |
|---|---|---|
| Average motion | 5.00 | 7.67 |
| Range | 2 | 3 |
| Variance | 0.67 | 1.56 |
| Stability | Higher | Lower |

Method B produces a higher average motion value, which indicates that it estimates greater motion. However, its larger range and variance show that its estimates fluctuate more between frames.

Method A has lower variance and a smaller range. Therefore, its motion estimation is more consistent over time.

**Conclusion**

Method A is more stable and consistent because it has a lower variance (0.67) and smaller range (2). Method B has a higher average motion (7.67) but greater variability. Hence, if the main objective is stable and reliable optical-flow estimation, Method A is preferable.

# 9. Detection Accuracy Comparison

Given results:

| Model | Correct Detections | Total Detections |
|---|---|---|
| A | 180 | 200 |
| B | 170 | 200 |

Detection accuracy measures the percentage of predictions that are correct out of the total number of predictions.

**(a) Compute Accuracy**

The formula is:

$Accuracy{=}Total\ PredictionsCorrect\ Predictions{\times}100$

Model A

$AccuracyA{=}200180{\times}100{=}0.90{\times}100{=}90\%$

Therefore,

Model A accuracy = 90%

Model B

$AccuracyB{=}200170{\times}100{=}0.85{\times}100{=}85\%$

Therefore,

Model B accuracy = 85%

**(b) Comparison and Evaluation**

| Parameter | Model A | Model B |
|---|---|---|
| Correct detections | 180 | 170 |
| Incorrect detections | 20 | 30 |
| Total | 200 | 200 |
| Accuracy | 90% | 85% |

Model A correctly identifies 180 out of 200 cases, whereas Model B correctly identifies 170 out of 200 cases.

The difference in accuracy is:

$90\%{-}85\%{=}5\%$

Thus, Model A is 5 percentage points more accurate than Model B.

In terms of incorrect detections:

- Model A makes 20 incorrect detections.
- Model B makes 30 incorrect detections.

Therefore, Model A makes 10 fewer errors for the same number of test cases.

**Critical Evaluation**

The results clearly indicate that Model A provides better detection performance on this dataset. Its 90% accuracy means that 9 out of every 10 detections are correct on average. Model B achieves 85% accuracy, which is still reasonable but lower than Model A.

However, accuracy alone does not provide information about precision, recall, false positives, or false negatives. Therefore, for a complete computer vision evaluation, additional metrics should also be considered. Based strictly on the given results, however, Model A is the stronger model.

**Conclusion**

Model A performs better than Model B, achieving 90% accuracy compared with 85%. It correctly detects 180 of the 200 cases and produces fewer errors. Hence, Model A is more reliable for detection on the given dataset.